

SMARTOPS AI ARCHITECTURE

System Overview Architecture

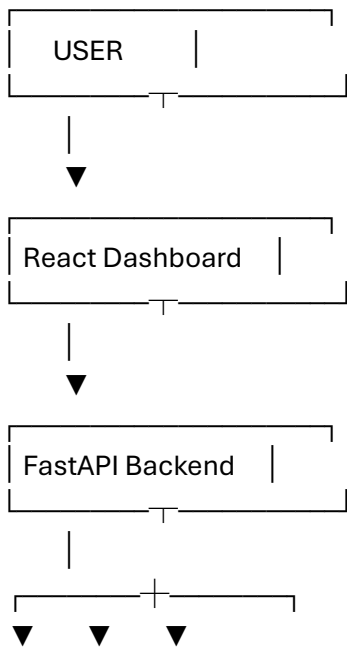
Introduction

SmartOps AI is an Agentic Incident Intelligence Platform designed to automate incident investigation and Root Cause Analysis (RCA). The system integrates runtime logs, GitHub commits, historical incidents, and Generative AI to reduce manual troubleshooting effort.

Architecture Goals

- Automated Incident Investigation
- Intelligent Root Cause Detection
- Historical Knowledge Reuse
- Faster Resolution Time
- AI-Powered Recommendations

High-Level Architecture



GitHub Logs Incident
Agent Agent Agent



Investigation Engine



RAG Agent



Gemini Agent



Risk Engine



Dashboard + PDF

Core Components

GitHub Agent

Responsible for:

- Fetching commits
- Fetching changed files
- Analyzing deployment history
- Tracking code modifications

Log Agent

Responsible for:

- Collecting runtime logs
- Parsing errors
- Identifying failures
- Monitoring application status

Incident Agent

Responsible for:

- Severity determination
- Service impact identification
- Business impact analysis

Internal Agent Architecture

Agent-Based Design

SmartOps AI follows an Agentic AI architecture where each agent performs a specialized task.

GitHub Agent



Commit Analysis

Log Agent



Error Extraction

Incident Agent



Impact Analysis



Investigation Engine



RAG Agent

▼
Gemini Agent

▼
Risk Engine

Investigation Engine

The Investigation Engine acts as the central orchestrator.

Responsibilities

- Correlate commits and logs
- Create event timeline
- Identify failure sequence
- Detect suspicious code changes

Example:

Deployment
|
▼
Config Change
|
▼
DB Timeout
|
▼
Payment Failure

RAG Architecture

The Retrieval-Augmented Generation module retrieves previous incidents.

Incident Query
|
▼
Embeddings
|
▼
Vector Search

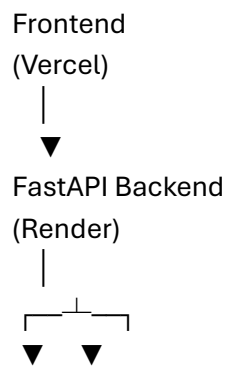


Benefits:

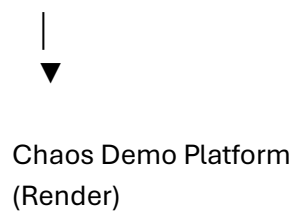
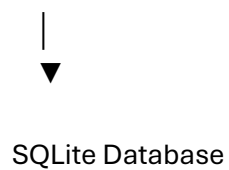
- Reuse past fixes
- Reduce investigation effort
- Improve AI accuracy

Deployment Architecture

Infrastructure Architecture



GitHub API
Gemini API



Database Structure

runtime_logs

Field	Description
id	Log ID
level	ERROR/WARN
message	Log Message
timestamp	Event Time

failure_events

Field	Description
id	Failure ID
type	DB Timeout
severity	Critical
timestamp	Event Time

Expected Outcome

Before SmartOps AI:

- RCA Time: 1–2 Hours

After SmartOps AI:

- RCA Time: 2–5 Minutes

Benefits:

- Faster RCA
- Reduced Downtime
- Better Reliability
- Historical Learning